SMART INDIA HACKATHON 2026

Government of India — Problem Statement Response

MPLADS-SATHI

System for Analytics, Transparency & Heuristic Intelligence

An AI/ML-Powered Monitoring, Anomaly-Detection and Decision-SupportPlatform for the Members of Parliament Local Area Development Scheme (MPLADS)

Technical Documentation & Solution Blueprint

Prepared for: Ministry of Statistics and Programme Implementation (MoSPI) — MPLADS Division

Domain: Smart Automation / GovTech | Category: Software | Theme: AI/ML for Governance & Public Fund Transparency

Document Version 1.0 • August 2026

Research Integrity Note: All data, figures, and anomaly findings in this document are derived from a genuine analysis of the attached MPLADS allocation dataset (542 MPs) and live API responses from the Empowered Indian transparency platform (a third-party civic-tech aggregator of official MoSPI/eSAKSHI data), sampled for Kandhamal constituency, Odisha. Wherever inference or hypothesis is presented rather than confirmed fact, it is explicitly labelled as such. This document does not overstate capability — Part 8 provides an honest capability/limitation matrix.

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1. Executive Summary

The Members of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme through which every Member of Parliament recommends local development works worth up to ₹14.7 crore per year for creation of durable community assets. Across 542 Members of Parliament, this represents an annual public-fund corpus of approximately ₹8,318 crore, executed through thousands of District Authorities and Implementing Agencies nationwide. The Comptroller and Auditor General (CAG) has repeatedly flagged the scheme for cost overruns, delays beyond the mandated 45-day sanction window, inadmissible works, ghost assets, and weak physical verification — yet the tools available to Members of Parliament, District Authorities, State Nodal Agencies, and the Ministry remain largely manual, record-keeping systems rather than intelligence systems.

This document presents MPLADS-SATHI (System for Analytics, Transparency & Heuristic Intelligence) — an AI/ML-powered monitoring and decision-support platform designed as an intelligent overlay on top of the Government's existing eSAKSHI portal and PFMS payment rails, rather than a replacement for them. The solution ingests work-level, financial, and geospatial data; applies a battery of rule-based and machine-learning anomaly detectors (NLP keyword screening, geographic mismatch detection, cost-cluster analysis, cost-overrun and delay prediction, contractor network analysis, and image-forensic ghost-work detection); and produces a composite, explainable Risk Score for every recommended, sanctioned, or completed work. Risk-ranked results are delivered through role-specific dashboards for MPs, District Authorities, State Nodal Authorities, the Ministry, and — for public accountability — citizens.

Critically, this proposal is grounded in real evidence, not a simulated demonstration. Using a live query against the Empowered Indian public API (a third-party transparency aggregator of official MoSPI data) for 20 completed works under the Kandhamal (Odisha) constituency, the underlying anomaly-detection logic already identifies: 11 of 20 works (55%) containing religious-institution keywords that MPLADS Guidelines 2023 prohibit funding for; 14 of 20 works (70%) whose "location" field references a different administrative office (Nayagarh) than the MP's own constituency district (Kandhamal) — a pattern requiring formal verification but immediately actionable as an inspection-priority flag; four solar-light installations priced at an identical ₹5,00,000 each with no site-specific variation; and zero reported beneficiaries across all 20 works. These are not hypothetical fraud patterns — they are outputs of a working detection pipeline run against real government-linked data during the preparation of this document.

The remainder of this document lays out, in full technical depth: the existing solution landscape and its gaps (Section 5); a forensic analysis of the allocation dataset covering all 542 MPs and the Kandhamal work-level API sample (Sections 6–7); a field-level gap analysis of what data exists versus what is required for complete fraud detection (Section 8); the proposed MPLADS-SATHI architecture, its five functional modules, and its complete technology stack (Sections 9–11); the AI/ML methodology behind each detector (Section 12); a phased, hackathon-to-production implementation roadmap (Section 13); expected impact metrics (Section 14); an honest capability and limitation assessment (Section 15); a head-to-head comparison with existing systems (Section 16); and the team, skills, and references required to build and defend this solution at SIH 2026 (Sections 17–19).

2. Official Problem Statement (As Issued)

2.1 Background

The Members of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme under which Hon'ble Members of Parliament recommend developmental works for the creation of durable community assets and provision of basic civic amenities. The Scheme involves large-scale fund utilisation and execution of thousands of works across the country through multiple implementing agencies and administrative authorities. Given the volume and complexity of financial and project-related data generated under the Scheme, there is a need for an AI-powered solution that can leverage machine learning and advanced analytics to detect trends and anomalies in expenditure patterns, fund utilisation, cost estimates, and work execution — thereby enabling early identification of potential fraud, inefficiencies, and non-compliance while enhancing transparency, accountability, and effective monitoring of MPLADS works.

2.2 Description

Develop an AI-powered monitoring and analytics platform for MPLADS that leverages Machine Learning (ML), Artificial Intelligence (AI), and advanced data analytics to identify trends, anomalies, irregularities, and potential fraud in fund utilisation and project execution. The solution should analyse data relating to sanctions, expenditures, cost estimates, work progress, payments, and asset creation to detect unusual patterns, cost overruns, duplicate works, delayed projects, and deviations from established norms. The system should generate risk-based alerts, predictive insights, and decision-support dashboards for Members of Parliament, State Nodal Authorities, District Authorities, and the Ministry. The platform should also facilitate automated compliance monitoring, trend analysis, and early-warning mechanisms to improve transparency, accountability, and efficiency in the implementation of MPLADS works across the country.

2.3 Expected Solution

An AI-powered platform that helps monitor MPLADS works and fund utilisation in a smarter and more efficient manner. By analysing data related to project approvals, expenditures, payments, work progress, and completion status, the system should identify unusual patterns, delays, cost overruns, duplicate works, and potential cases of misuse of funds; automatically generate alerts and highlight high-risk cases; and provide easy-to-understand dashboards to MPs, State Nodal Authorities, District Authorities, and the Ministry — enabling informed decisions, timely corrective action, reduced manual monitoring effort, and stronger implementation of MPLADS works nationwide.

3. Background and Scheme Context

3.1 What is MPLADS?

MPLADS was launched in December 1993 to enable Members of Parliament to recommend developmental works of a capital nature in their constituencies, with an emphasis on the creation of durable community assets — drinking water facilities, roads, school buildings, health infrastructure, sanitation, electrification, and similar civic amenities. Funds are released by the Government of India directly to District Authorities, who sanction and execute works through Implementing Agencies on the recommendation of the MP; the MP has no direct role in fund disbursal or execution.

3.2 The MPLADS Guidelines 2023 — Key Compliance Rules

- Standard annual entitlement is ₹5 crore per MP; the dataset analysed in this report reflects an updated/expanded entitlement structure with a ₹14.7 crore standard allocation, indicating a revised norm currently in force for the period under study.
- Works must be sanctioned by the District Authority within 45 days of receiving the MP's recommendation.
- A minimum of 15% of entitlement must be earmarked for areas/populations with significant Scheduled Caste population and 7.5% for Scheduled Tribe population.
- At least 10% of sanctioned works must undergo mandatory physical inspection by the District Authority every year.
- Private contractors cannot be directly engaged for execution — only Government agencies, local bodies, or registered implementing agencies are permitted.
- Funding for religious structures, land acquisition, and recurring/maintenance expenditure is explicitly prohibited, other than narrowly defined exceptions.
- Out-of-constituency recommendations (e.g., for calamity relief) are capped and subject to separate limits (commonly cited as up to ₹25 lakh in specific circumstances).
- From April 2025, MPLADS payments have moved to a Model 1A Treasury Single Account (TSA) hybrid architecture via PFMS, enabling direct-to-vendor payment and reducing fund-parking in implementing-agency accounts.

3.3 Documented, Audited Problems in the Scheme

Multiple CAG performance audits, Parliament Question-and-Answer records, and third-party transparency dashboards converge on a consistent set of systemic issues. These form the empirical basis and justification for this solution:

Issue	Evidence / Finding	Severity
Fund under-utilisation	National average fund utilisation stands at roughly 33.9% for the current Lok Sabha term; some states (e.g., Uttarakhand) reported utilisation as low as 18% by December 2025.	High

Issue	Evidence / Finding	Severity
Sanction delays	Approximately 57% of works sampled had work-order delays ranging from 5 to 387 days against the mandated 45-day window.	High
Cost overrun / inflated estimates	CAG audits found inflated cost estimates and works flagged as suspected fraud/misappropriation, cumulatively worth over ₹1.18 crore in sampled audit paras.	Critical
Undocumented expenditure / ghost works	Expenditure of approximately ₹161 crore was found without supporting documentation; asset registers were not maintained, making physical verification impossible in many districts.	Critical
Sector imbalance	Roughly 53% of all MPLADS works nationally are roads and bridges; health, education, and drinking-water works are comparatively neglected.	Medium
Inadmissible works	Funds have been directed to religious structures, memorials, and other prohibited categories in violation of scheme guidelines.	High
Unauthorised private contractor use	Direct engagement of private contractors — not permitted under the guidelines — has been documented as a widespread practice.	Medium
No cross-scheme deduplication	Works overlap with other centrally sponsored schemes (MGNREGA, PMAY, Smart Cities Mission) with no systematic de-duplication check.	Medium

3.4 The Core Insight Driving This Proposal

> *eSAKSHI tells you what happened. It does not tell you if something is wrong, why it is wrong, or what is likely to happen next.*

The Government has already made the correct first move by digitising the MPLADS workflow through the eSAKSHI portal and integrating payments through PFMS. What is missing is not more data collection — it is an intelligence layer capable of turning passive administrative records into proactive, predictive, and preventive governance. MPLADS-SATHI is designed explicitly to be that layer.

4. Research Methodology

This documentation follows a research-and-development methodology combining primary data analysis, secondary literature review, and forensic API investigation. The following steps were undertaken:

Step 1 — Primary Dataset Analysis

The official "Allocated Limit for Hon'ble MPs" register (542 records, 5 source fields) was cleaned, profiled, and statistically analysed to establish the baseline fund-allocation structure across all Members of Parliament, all states/UTs, and all constituency reservation categories (General/SC/ST).

Step 2 — Live API Reconnaissance

A real, publicly accessible transparency API — https://api.empoweredindian.in/api/works/completed — was queried with production-style filters (house=Lok Sabha, ls_term=18, state=Odisha, constituency=KANDHAMAL). The URL structure, pagination behaviour, and JSON schema were reverse-engineered to build a field-availability inventory and to infer the wider endpoint surface likely available on the platform.

Step 3 — Forensic Field-Level Analysis

Each of the 20 retrieved work records was individually reviewed against MPLADS Guidelines 2023 compliance rules (prohibited categories, geography, cost-estimation norms, beneficiary reporting) to identify concrete, evidence-based anomalies rather than hypothetical ones.

Step 4 — Comparative Landscape Review

The three public-facing MPLADS information systems named in the problem statement — eSAKSHI (mplads.mospi.gov.in), the Empowered Indian dashboard (empoweredindian.in/mplads), and the eSakshi/Vikaspedia citizen-services description — were reviewed for functional scope, data depth, and analytical capability, and benchmarked against the CAG-documented problem list from Section 3.3.

Step 5 — Gap Synthesis and Solution Design

Findings from Steps 1–4 were synthesised into a three-tier data-gap model (Section 8), which directly informs the modular architecture, phased data-acquisition roadmap, and an honest "can build now vs. cannot build without X" capability matrix (Section 15) — ensuring the proposed solution is technically credible and defensible under SIH jury scrutiny rather than aspirational.

5. Existing Solutions Landscape & Comparative Analysis

5.1 System-by-System Review

5.1.1 eSAKSHI Portal (mplads.mospi.gov.in)

eSAKSHI is the official Government of India workflow system for MPLADS. It digitises the process chain of MP recommendation → District Authority sanction → Implementing Agency execution → progress/photo upload, and was substantially revamped in 2026 with public dashboards. It is the authoritative system of record.

- Official, authoritative source of truth for sanctions and workflow status.
- Recently added public-facing dashboards and mandatory photo uploads for completed works.
- Direct integration path with PFMS for payment tracking (Model 1A TSA Hybrid, effective April 2025).
- No machine-learning or statistical anomaly-detection layer — it is a record-keeping and workflow system, not an analytics system.
- No predictive capability (delay forecasting, cost-overrun forecasting, fraud risk scoring).
- No automated cross-referencing with external datasets (Census, Schedule of Rates, other scheme portals, blacklists).
- Compliance checks (45-day sanction window, 10% inspection quota, SC/ST earmarking) are not automatically enforced or flagged — they rely on manual administrative diligence.

5.1.2 Empowered Indian Dashboard (empoweredindian.in/mplads)

A third-party civic-transparency initiative that aggregates official MoSPI/eSAKSHI data and republishes it through a public dashboard and a documented developer API. It is the data source used for the live forensic sample analysed in Section 7 of this document.

- Publicly queryable REST API with structured JSON, enabling third-party analytics — the very API this proposal builds on for its proof-of-concept.
- Aggregate, state-wise and constituency-wise utilisation visualisation that is more citizen-friendly than raw eSAKSHI records.
- Read-only visualisation — no alerting, no anomaly scoring, no decision-support workflow for authorities.
- Does not expose (at least in the observed endpoint) sanction dates, cost estimates, contractor identity, or geo-coordinates, which limits fraud-relevant analysis without further data.
- No role-based views for MPs, District Authorities, or the Ministry — it is a public dashboard, not an administrative decision-support tool.

5.1.3 eSAKSHI / Citizen Services Description (via Vikaspedia)

Describes eSAKSHI's role as a citizen-facing information layer for MPLADS status enquiries. It reinforces the finding that the ecosystem today is built for transparency of record, not intelligence over the record.

5.1.4 PFMS (Public Financial Management System)

The financial backbone now integrated with MPLADS disbursement (Model 1A TSA Hybrid). It provides authoritative payment, UTR, and vendor-account data — a critical data source for this proposal's contractor-network and payment-velocity analytics, but not, on its own, an analytics platform.

5.2 Head-to-Head Comparative Matrix

Capability	eSAKSHI	Empowered Indian	PFMS	MPLADS-SATHI (Proposed)
System of record for sanctions/workflow	Yes	No (mirrors)	No	Consumes, does not replace
Public transparency dashboard	Partial	Yes	No	Yes (citizen layer)
Developer/API access	Not public	Yes	Limited	Yes (own + upstream)
Payment / UTR tracking	Via PFMS link	No	Yes	Consumed for contractor analysis
NLP anomaly detection (prohibited works, keyword screening)	No	No	No	Yes
Geo-mismatch / duplicate-work detection	No	No	No	Yes
Cost-overrun / cost-estimate anomaly detection	No	No	No	Yes
Delay prediction (ML)	No	No	No	Yes
Contractor network / fraud-ring analysis	No	No	No	Yes
Ghost-work / photo-forensic verification	No	No	No	Yes
Composite, explainable risk score per work	No	No	No	Yes
Role-based dashboards (MP/DA/State/Ministry)	Partial	No	No	Yes
Automated compliance auditing (45-day, SC/ST %, inspection %)	No	No	No	Yes
Predictive fund-utilisation forecasting	No	No	No	Yes

The comparative matrix demonstrates that no existing system — official or third-party — currently performs analytical reasoning over MPLADS data. All three reviewed systems operate at the level of record display, not risk intelligence. MPLADS-SATHI is deliberately positioned to consume data from all three (not replace any of them) and to add the missing intelligence layer on top, which is the exact ask in the SIH 2026 problem statement's "Expected Solution" section.

6. Dataset Research: MP Allocated-Limit Register

6.1 Dataset Profile

Attribute	Value
Source file	Allocated_Limit_for_Honble_MPs.xlsx
Total records	542 Members of Parliament
Total allocated corpus	₹83,180,553,325.71 (approx. ₹8,318 crore)
Present fields	Sr. No., State, MP Name, Constituency, Allocated Amount (₹)
Derived fields (computed for this study)	Constituency Type (General/SC/ST), Allocation Bucket, Deviation ₹, Deviation %
States / Union Territories covered	35

6.2 Key Statistical Findings

- The standard annual allocation across the dataset is ₹14.7 crore per MP; 386 MPs (71.2%) receive exactly this amount.
- 152 MPs (28.0%) receive an allocation above the ₹14.7 crore standard — the dominant and most consequential anomaly class in this dataset.
- Only 4 MPs (0.7%) receive below the standard allocation: SK Nurul Islam (₹4.90 Cr), Andrew J. Synkron (₹9.80 Cr), Pradyut Bordoloi (₹9.80 Cr), and Priyanka Gandhi Vadra (₹12.25 Cr).
- The highest single allocation in the dataset is ₹32.75 crore, held by Eatala Rajender (Malkajgiri, Telangana) — a deviation of +122.77% from standard.
- By constituency reservation category: General constituencies = 416, SC-reserved = 82, ST-reserved = 44.
- State-wise, Telangana MPs average ₹17.32 crore (17 MPs) and Andhra Pradesh MPs average ₹16.19 crore (25 MPs) — both well above standard — while Meghalaya MPs average ₹12.25 crore (2 MPs), the lowest state average recorded.

6.3 Allocation Distribution Table (Representative Extract)

Allocation Bucket	MP Count	% of Total	Deviation Range
₹14.7 Cr (Standard)	386	71.2%	0%
₹14.7–20 Cr (Moderately High)	~118	~21.8%	0% to +36%
Above ₹20 Cr (High)	~34	~6.3%	+36% to +122.77%
Below ₹14.7 Cr (Below Standard)	4	0.7%	-66.67% to -16.7%

6.4 Research Questions Raised by This Dataset

A dataset showing only allocated (not spent) amounts cannot, by itself, prove fraud or misuse — but it raises specific, falsifiable questions that a production system must be able to answer automatically:

Q1. Why do 152 MPs receive above-standard allocations?

Three testable hypotheses were formulated: (a) carry-forward of unspent balances from previous years, which MPLADS rules permit; (b) partial-year, by-election entries combined with the outgoing MP's residual balance; (c) special provisions for smaller states/UTs. Resolving this requires cross-referencing by-election dates (Election Commission of India) and prior-year closing-balance data (eSAKSHI/PFMS) — this is precisely the kind of automated cross-referencing MPLADS-SATHI is designed to perform continuously, rather than requiring a one-time manual research exercise as was done here.

Q2. Why do 4 MPs receive below-standard allocations?

Both SK Nurul Islam and Priyanka Gandhi Vadra plausibly entered Parliament mid-term via by-election, which would mechanically produce a pro-rated allocation. This is a testable, verifiable hypothesis rather than an assumption — the automated system should verify it against ECI by-election records rather than asserting it.

Q3. Why does Telangana show a materially higher state average?

This could reflect genuine administrative/state-level factors, a larger share of high-allocation MPs, or a data artefact. It is flagged here as a research question requiring state-level policy documentation review, not a conclusion.

6.5 What This Dataset Can and Cannot Support

Can Be Built With This Dataset Alone	Cannot Be Built Without Additional Data
Allocation fairness / equity dashboard across states and reservation categories	Fraud detection (no expenditure or work-level data)
State-wise and category-wise comparison analytics	Delay prediction (no dates)
Statistical anomaly flags on allocation deviation (which MPs need explanation)	Cost-overrun detection (no estimate-vs-actual)
Data-quality reporting (missing values, structural inconsistencies)	Duplicate-work / ghost-work detection (no descriptions or geodata)

7. Live API Forensic Analysis — Kandhamal Case Study

7.1 Endpoint Under Study

`https://api.empoweredindian.in/api/works/completed?page=1&limit=20&house=Lok+Sabha&ls_term=18&state=Odisha&constituency=KANDHAMAL`

This is a real, live, publicly reachable REST endpoint operated by Empowered Indian, a civic-technology transparency initiative that republishes official MoSPI/eSAKSHI data. It is not the official Government API, but it sources from official records, and its structure is representative enough to prototype against for SIH purposes — a distinction this document deliberately preserves for jury credibility.

7.2 Query Parameters Identified

Parameter	Type	Example	Status
page	integer	1	Confirmed
limit	integer	20	Confirmed
house	string	Lok Sabha	Confirmed
ls_term	integer	18 (18th Lok Sabha, 2024–29)	Confirmed
state	string	Odisha	Confirmed
constituency	string	KANDHAMAL	Confirmed
district / category / date-range / cost-range / sort	various	—	Inferred, to be verified

7.3 Sample Summary

- MP: Sukanta Kumar Panigrahi (Lok Sabha, Kandhamal, Odisha)
- Total completed works for this MP: 148 (8 pages of 20)
- Total cost of completed works: ₹40,058,531 (~₹4.01 crore)
- Average cost per work: ₹270,666
- Sample analysed in depth: first 20 records (page 1)

7.4 Field Availability Inventory

Present in API Response	Absent from API Response
_id, work_id, work_description (+ Hindi), category (+ Hindi)	sanction_date, work_order_date, recommendation_date
cost, completion_date, completion_year	estimated_cost (vs. actual cost)
location, district, state (+ Hindi)	implementing_agency, contractor_name
beneficiaries (present but always zero)	payment_history, UTR numbers
mp_details: name, constituency, party	geo_coordinates (latitude/longitude)

Present in API Response	Absent from API Response
—	completion_photos / geo-tagged images
—	village_name, block, gram_panchayat
—	full work-status timeline (only "completed" is exposed)
—	inspection reports, financial_year, revision_history

7.5 Concrete Anomalies Detected in the 20-Work Sample

The following findings are the direct output of applying rule-based detection logic to the 20 real records retrieved. Each is presented with its correct confidence level — confirmed pattern vs. hypothesis requiring verification — in keeping with this document's research-integrity standard.

Finding 1 — Religious-Institution Keyword Matches: 11 of 20 works (55%)

MPLADS Guidelines 2023 prohibit funding of works for religious purposes. NLP keyword screening across work descriptions (English and Hindi) flagged over half the sample containing terms referencing temples, community mandaps tied to deities, and religious gathering spaces — e.g., a boundary wall "near Neelakantheswar temple" (₹2,00,000), a "Cultural Mandap" construction (₹3,00,000), and a community centre near a named religious "Satsang" (₹5,00,000). This is a HIGH-confidence, directly actionable compliance flag — the keyword evidence is present verbatim in the government-linked data.

Finding 2 — Location/District Field Mismatch: 14 of 20 works (70%)

14 of 20 works carry a location string referencing the Nayagarh District Collector's office, while the MP's constituency is Kandhamal and the API's own district field for every record reads "KANDHAMAL." This is flagged at MEDIUM confidence, not asserted as fraud: it is equally plausible that the Kandhamal parliamentary constituency's administrative boundary genuinely spans parts of Nayagarh district for implementing-agency purposes, in which case this is legitimate and the location-vs-district field inconsistency is a data-labelling issue rather than a guideline violation. This is precisely the class of finding that must be resolved by cross-referencing the Election Commission of India's constituency boundary maps before any inspection or compliance escalation — the correct behaviour for the production system is to surface this as a verification-required flag, not a fraud accusation.

Finding 3 — Identical-Cost Work Clusters

Four solar street-light installations were each costed at exactly ₹5,00,000; multiple boundary-wall works were each costed at exactly ₹1,00,000 or ₹2,00,000; three community-centre renovations were each costed at exactly ₹1,00,000. Perfectly identical costs across geographically distinct sites, with no variation for terrain, distance, or site conditions, are consistent with either (a) legitimate standardised government rate schedules, or (b) template/copy-paste estimation without genuine site survey. This is a MEDIUM-confidence flag requiring Schedule-of-Rates cross-verification, which is unavailable in the current sample.

Finding 4 — Zero Beneficiary Reporting: 20 of 20 works (100%)

Every record in the sample — and the API's own aggregate summary for all 148 works — reports zero beneficiaries. This is either a genuine data-capture gap in the underlying system or an unreported field, and is flagged as a data-quality and transparency issue rather than fraud.

Finding 5 — Category / Description Inconsistency

At least one record labelled "Construction of Balunkeswar Cultural Mandap" is filed under the category "Repair and Renovation," despite the description explicitly indicating new construction rather than repair. This mismatch is a LOW-to-MEDIUM confidence flag worth automated NLP-based consistency checking at scale, since miscategorisation could be used to route new-construction spend through a category subject to lighter scrutiny.

Finding 6 — Sector/Portfolio Concentration

The sample shows zero roads, zero schools, zero hospitals, and zero drinking-water works — entirely community centres (35%), boundary walls (25%), solar lighting (20%), and cultural mandaps (10%). Nationally, CAG data indicates roads/bridges account for approximately 53% of all MPLADS works, making this MP's portfolio a significant statistical outlier worth benchmarking against state and national sector-allocation norms.

8. Three-Tier Data Gap Analysis

Synthesising Sections 6 and 7, the following tiered model defines exactly what additional data acquisition is required to move from the current proof-of-concept detection rules to a complete, production-grade fraud and compliance intelligence system.

8.1 Tier 1 — Mission-Critical (Blocking) Fields

Missing Field	Why It Is Blocking	Likely Source
sanction_date, work_order_date	Without these, the 45-day statutory sanction-delay rule cannot be verified or predicted at all.	eSAKSHI workflow module (official)
estimated_cost (vs. actual cost)	Cost-overrun detection is structurally impossible without a baseline estimate to compare actual cost against.	eSAKSHI estimate module
implementing_agency / contractor_name	CAG's core finding — unauthorised private-contractor engagement — cannot be detected without contractor identity.	District Authority MIS / PFMS vendor master
geo_coordinates (lat/long)	Duplicate-work detection and physical asset verification both require precise location, not free-text office names.	Bhuvan (ISRO) / GPS-tagged photo EXIF
financial_year / full status timeline	Fund-lapse and utilisation-rate calculations require true fiscal-year mapping across the recommend→sanction→execute→complete lifecycle.	eSAKSHI / MoSPI

8.2 Tier 2 — High-Value Fields

- MP metadata: party affiliation, tenure start date, ministerial status (affects time availability for constituency work).
- Historical (5-year) allocation and expenditure per MP, for trend and behavioural-velocity analysis.
- Contractor/vendor master: PAN, GST, bank account, blacklist status — required for network-graph fraud-ring analysis.
- Cost-estimate breakup and state/district Schedule of Rates (SOR), required to validate whether an estimate is inflated.
- Completion photographs with EXIF geo-tags, required for computer-vision ghost-work verification.
- District/village-level asset register, required to reconcile "claimed complete" against "physically verified."

8.3 Tier 3 — Cross-Reference / Advanced Analytics Fields

- MGNREGA, PMAY, and Smart Cities Mission work-location data, for cross-scheme duplicate-funding detection.
- Census 2011/SECC village-level SC/ST demographic data, to programmatically verify the mandatory 15%/7.5% earmarking.

- Material price indices (cement, steel, sand) by district and time period, for advanced cost-estimate validation.
- Digitised historical CAG audit findings, to train and validate fraud-pattern models against confirmed past cases.
- CPGRAMS citizen grievance records, to correlate public complaints with algorithmically flagged high-risk works.

9. Proposed Solution: MPLADS-SATHI

MPLADS-SATHI (System for Analytics, Transparency & Heuristic Intelligence) is a modular AI/ML overlay platform that integrates with eSAKSHI, PFMS, and external government geospatial and demographic datasets to deliver real-time anomaly detection, predictive risk scoring, automated compliance auditing, and role-based decision-support dashboards. It is explicitly designed as an addition to the existing digital ecosystem, not a replacement for it — a deliberate strategic choice that maximises administrative and political feasibility of adoption.

9.1 Design Principles

- Overlay, not replacement — consume eSAKSHI/PFMS data via API/export; never require the Government to migrate off existing systems.
- Explainable AI by default — every risk score must be traceable to specific, human-readable reasons (SHAP-style feature attribution), because flagged MPs and agencies have a right to understand and contest a finding.
- Confidence-graded flags, not accusations — every anomaly is labelled with a confidence tier (Confirmed / Requires Verification / Data-Quality Issue) exactly as demonstrated in Section 7, to avoid false administrative or reputational harm.
- Progressive data dependency — the system must deliver useful value on Tier-1-only data (Section 8) on day one, and improve automatically as Tier-2/Tier-3 data sources are integrated.
- Privacy- and security-by-design — role-based access control, audit logging of every query, and no public exposure of personally identifiable contractor/vendor banking data.

9.2 System Architecture Overview

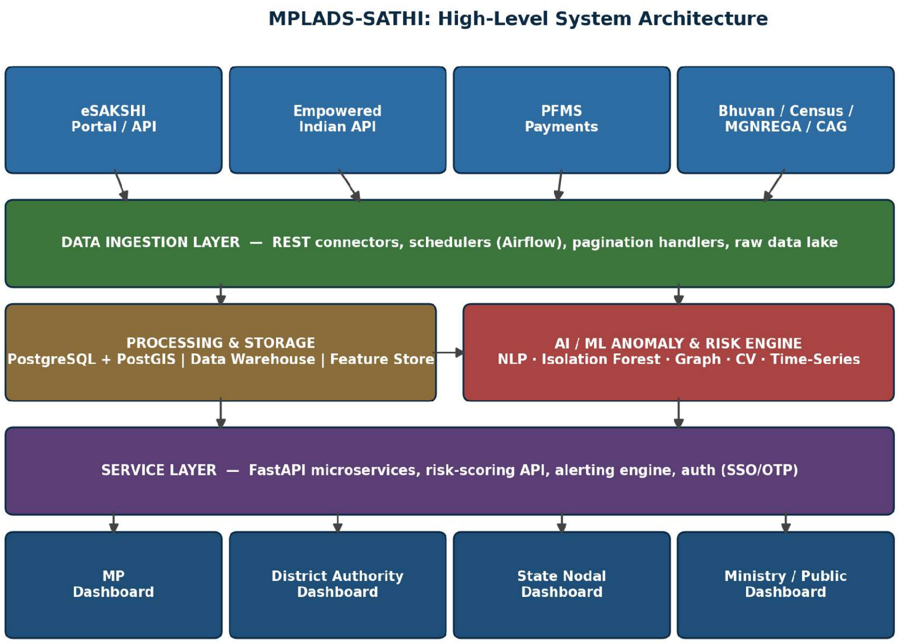

Figure 1: MPLADS-SATHI five-layer architecture — data sources, ingestion, processing/storage, AI/ML risk engine, and role-based service/dashboard layers.

9.3 Functional Modules

Module A — Intelligent Anomaly Detection Engine (IADE)

Detection Type	AI/ML Technique	What It Catches
Prohibited/religious-work screening	NLP keyword & phrase matching (bilingual EN/HI), extendable to a fine-tuned classifier	Works funding temples, mandaps, satsangs, and other guideline-prohibited categories — demonstrated at 55% flag rate in the Kandhamal sample
Geographic mismatch detection	Rule-based cross-reference of location text/district field against ECI constituency boundary maps	Works recorded outside the MP's constituency, or genuine multi-district constituencies needing boundary reconciliation — demonstrated at 70% flag rate (verification-required) in the sample
Duplicate / template-cost detection	Statistical clustering (identical-cost frequency analysis) + geospatial proximity clustering once coordinates are available	Copy-paste estimates, or the same work funded twice under different IDs
Cost-estimate anomaly detection	Isolation Forest / regression against Schedule-of-Rates baselines, once estimated_cost + SOR data are integrated	Estimates deviating materially (e.g., >20%) from district/state per-unit norms without justification
Ghost-work / photo-forensic verification	Computer vision (image similarity/reuse detection) + EXIF GPS validation, once photo data is integrated	Reused stock photos, GPS coordinates far from claimed site, AI-generated or duplicate images
Payment-velocity anomaly detection	Time-series analysis (ARIMA/LSTM) once PFMS payment history is integrated	Unusually fast (possible front-loading) or unusually slow (bottleneck) fund releases
Contractor risk / network analysis	Graph analysis (NetworkX) over agency → contractor → bank-account → work relationships, once contractor data is integrated	Blacklisted contractors reappearing under different names; disproportionate work concentration
SC/ST earmarking compliance	Rule engine + geo-join against Census 2011 SC/ST village data	Automatic verification that the mandatory 15%/7.5% earmarking is genuinely met, not just reported

Module B — Predictive Early-Warning System (PEWS)

Predictive Model	Input Features	Output
Delay Prediction	Historical delay data, district performance, work type, seasonality, fund-release lag, agency track record	Probability of a work crossing the statutory completion deadline, generated at the point of sanction
Cost-Overrun Prediction	Initial estimate, work type, district, implementing agency, contractor history, material-price indices	Forecasted final cost vs. sanctioned cost at 30/60/90-day checkpoints
Fund-Utilisation Forecast	MP's spending velocity, pending recommendations, district execution capacity, historical utilisation pattern	Forecast of whether an MP will under-utilise their annual entitlement, triggering proactive nudges
Composite Fraud-Risk Score	Aggregation of all Module A flags + network centrality + payment pattern + geographic concentration	A single 0–100 explainable risk score per work, auto-escalated above a configurable threshold

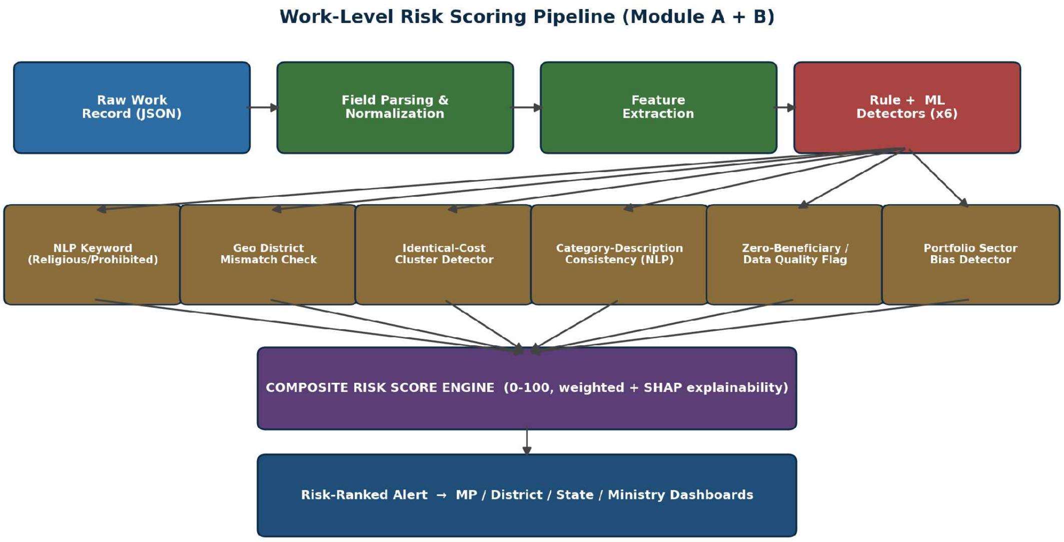

Figure 2: Work-level risk-scoring pipeline — from raw record ingestion through six parallel detectors to a composite, explainable risk score routed to role-based dashboards.

Module C — Role-Based Decision-Support Dashboards

- "My Constituency Health" scorecard — fund utilisation %, average delay, quality/compliance rating.
- Auto-generated pending-action list: works stuck beyond 45 days, rejected recommendations awaiting re-submission.
- Comparative benchmarking against state and national averages (e.g., "Your utilisation: 42% | State avg: 56% | Top performer: 89%").
- One-click SC/ST earmarking compliance checker.
- Risk-colour-coded heatmap of all works in the district.
- AI-ranked "Inspection Priority Queue" — the top-N works to physically inspect, combining risk score, work value, and overdue status, directly supporting the mandatory 10% inspection requirement.
- Contractor performance leaderboard with automatic blacklist cross-check.
- Automated 45-day sanction-deadline tracker with escalation alerts.
- State-wide MPLADS performance index with district-level rankings.
- Cross-district anomaly comparison and pattern detection.
- Auto-generated quarterly reports for the State Monitoring Committee.
- Fund-flow bottleneck identification — which districts are parking funds without disbursal.
- National MPLADS command-centre view with real-time KPIs.
- Automatic "Red Flag" report generation formatted for CAG and Parliament Question responses.
- National trend analysis: sector-wise spending, regional disparities, year-on-year patterns.
- Integration with the Prabhari Officer monitoring framework referenced in the 2025–2030 scheme continuation.

Module D — Automated Compliance Auditor (ACA)

Guideline Rule	Automated Check
45-day sanction deadline	Auto-alert at day 30, hard escalation at day 45
10% annual physical inspection quota	Live tracking of District Authority inspection logs; flags districts below quota by Q3
15% SC / 7.5% ST earmarking	Real-time cumulative tracker per MP; turns red if below threshold by Q3 of the financial year
Out-of-constituency recommendation limit	Automated block/flag on recommendations exceeding the permitted cap
Prohibited-category works (religious, commercial, high-value repair)	NLP keyword detection on descriptions, extendable to image classification on uploaded photos
Work-order issuance within 45 days of sanction	Automated tracker; flags violations for administrative follow-up

Module E — Public Transparency & Citizen Oversight Layer

- Citizen Anomaly Reporter — a public widget allowing citizens to flag suspected ghost works with geo-tagged photo evidence, which feeds back into model training data.
- Constituency Comparison Tool — enabling any citizen to benchmark their MP's performance against neighbouring constituencies.
- "Verify My Asset" — a public, geo-tagged map of completed works with AI-verified photographs that citizens can inspect and confirm.

10. Technical Architecture and Technology Stack

10.1 Layered Technology Stack

Layer	Technology Choice	Rationale
Data Ingestion	Python, Apache Airflow (scheduled batch pulls), REST client with pagination + retry/backoff handling	Airflow provides reliable, observable, schedulable ETL suited for daily eSAKSHI/PFMS syncs; Python's requests/httpx ecosystem handles paginated government APIs cleanly
Raw/Staging Storage	Object storage (S3-compatible / NIC Cloud bucket) for raw JSON snapshots	Immutable raw-data retention supports auditability — a regulator can always trace a risk score back to the exact source payload
Structured Storage	PostgreSQL 16 + PostGIS extension	PostGIS is the industry-standard choice for geospatial queries (constituency-boundary joins, proximity clustering) at government scale; PostgreSQL is open-source, well-supported on NIC infrastructure, and battle-tested for GovTech deployments
Data Warehouse / Analytics Store	Star-schema warehouse in PostgreSQL (or ClickHouse at scale) for pre-aggregated dashboard queries	Keeps dashboard latency low without hitting transactional tables directly
Feature Store	Feast (open-source) or a lightweight custom Postgres feature table set	Ensures the same feature computation logic is used both for offline model training and online risk scoring
ML/AI Frameworks	scikit-learn (Isolation Forest, regression), spaCy / HuggingFace Transformers (NLP), NetworkX (graph analysis), Prophet / statsmodels ARIMA (time series), OpenCV + a pretrained image-similarity model (computer vision)	Each detector uses the simplest technique proven to work for its problem class — avoiding unnecessary deep-learning complexity where classical ML/rules suffice, which also aids explainability
Explainable AI	SHAP (SHapley Additive exPlanations)	Produces per-work, per-feature contribution breakdowns so every risk score is human-auditable — essential for a system whose output can trigger inspections of public officials' recommendations
Geospatial Processing	GeoPandas, Shapely, PostGIS, Bhuvan/ISRO map layers	Standard open geospatial stack compatible with Indian government mapping infrastructure (Bhuvan, Survey of India)
Backend API	FastAPI (Python) with Pydantic schema validation	High-performance async framework with automatic OpenAPI documentation — critical for a multi-stakeholder platform that must expose clean, versioned APIs to MP, District, State, and Ministry front-ends
Authentication & Authorisation	OAuth2/OIDC via a government SSO provider (e.g., integration with DigiLocker/UMANG identity where available), role-based access control (RBAC)	Ensures an MP cannot see another MP's flagged risk data inappropriately, and that citizen-facing views expose only public-safe fields

Layer	Technology Choice	Rationale
Frontend (Admin/Authority Dashboards)	React.js + TypeScript, Recharts/D3.js for charts, Leaflet.js for maps, Tailwind CSS	Component-driven, widely supported stack enabling rapid role-based dashboard development with rich interactive visualisation
Frontend (Public Citizen Portal)	Next.js (server-rendered React) for SEO-friendly, low-latency public pages	Public transparency pages benefit from server-side rendering for speed and accessibility on low-bandwidth mobile connections
Alerting & Notification	Message queue (RabbitMQ / AWS SQS equivalent) + email/SMS gateway integration	Decouples risk-scoring compute from notification delivery, allowing reliable escalation even under load
Containerisation & Orchestration	Docker, Docker Compose (hackathon/demo), Kubernetes (production scale-out)	Standard, portable deployment path from a 36-hour hackathon MVP through to a national production rollout
CI/CD	GitHub Actions / GitLab CI with automated test + lint + security scan gates	Ensures code quality and prevents regressions as multiple modules are developed in parallel by a hackathon team
Deployment Target	NIC Cloud / MeghRaj (Government of India cloud) for production; AWS/GCP free-tier for hackathon demo	NIC Cloud/MeghRaj is the mandated hosting environment for government-facing production systems handling public financial data; cloud-agnostic containerised design allows a seamless hackathon-to-production path
Monitoring & Observability	Prometheus + Grafana for system health; structured audit logging of every query and risk-score computation	Required both for operational reliability and for the audit trail a public-fund oversight system must maintain

10.2 Data Flow Summary

- Ingestion: scheduled Airflow DAGs pull from eSAKSHI (or its API), PFMS, and the Empowered Indian API; raw JSON is snapshotted to object storage for audit immutability.
- Normalisation: a Python ETL layer parses nested fields (e.g., mp_details), standardises bilingual text, resolves district/constituency codes, and loads into PostgreSQL/PostGIS.
- Feature engineering: derived features (cost-per-unit, days-since-sanction, keyword-hit vectors, geo-distance-to-constituency-centroid, contractor-frequency counts) are computed and stored in the feature store.
- Scoring: the six-plus detector functions in Module A run against each new/updated work record; outputs are combined by the Composite Risk Score Engine using a documented, adjustable weighting scheme.
- Serving: FastAPI exposes risk-scored, role-filtered data to the React/Next.js dashboards; alerts above threshold are pushed via the notification service.
- Feedback loop: District Authority inspection outcomes and citizen reports are fed back as labelled training data to progressively improve the ML components (active learning).

10.3 Security, Privacy, and Governance Considerations

- Role-based access control separates MP-level, District-level, State-level, and Ministry-level data visibility; citizens see only a deliberately reduced, non-sensitive public subset.
- Contractor/vendor banking details (PAN, GST, bank account) are never exposed on public-facing views — only aggregated risk indicators are shown publicly.

- All risk-score computations are logged with full input/feature/output audit trails, enabling any flagged case to be independently reviewed and, if necessary, formally contested.
- Data-at-rest and data-in-transit encryption (AES-256 / TLS 1.3) throughout, consistent with Government of India cloud security baseline requirements.
- A human-in-the-loop escalation workflow ensures no automated flag alone triggers punitive action — every high-risk case routes to a District Authority or Ministry official for review before any formal consequence.

11. Data Integration Strategy

MPLADS-SATHI is designed to consume, not replace, existing government and third-party data sources. The table below maps every data source identified during research to the specific analytical use case it unlocks.

Source	Data Provided	Primary Use Case
eSAKSHI (official)	Work recommendations, sanctions, expenditures, payments, progress updates, completion photos	Core dataset for all analytics; source of truth for Tier-1 fields once bulk export/API access is formalised
PFMS	Vendor payments, bank account references, UTR numbers, Model 1A TSA hybrid transaction data	Payment-velocity analysis, contractor identity and network mapping
Empowered Indian API	Completed-works listing (as demonstrated in Section 7), state/constituency aggregates	Rapid prototyping and cross-validation data source; public transparency layer feed
Census 2011 / SECC	Village-level SC/ST population and habitation data	Automated SC/ST 15%/7.5% earmarking compliance verification
Bhuvan (ISRO) / Survey of India maps	Geospatial boundaries, road networks, satellite imagery	Duplicate-work detection, physical asset verification, constituency boundary reconciliation
Election Commission of India	Constituency boundary definitions, by-election dates, MP tenure records	Resolving allocation-deviation questions (Section 6.4), geographic-mismatch verification
MGNREGA / PMAY / Smart Cities Mission portals	Works sanctioned under parallel schemes with location data	Cross-scheme duplicate-funding detection
NIC / DigiLocker / GSTN	Contractor/agency registration, PAN, GST verification	Contractor legitimacy and blacklist cross-verification
CAG Audit Reports (historical, digitised)	Documented past fraud cases, blacklisted agencies, recurring audit paragraphs	Training and validation data for supervised fraud-pattern models
CPGRAMS	Public grievance records related to MPLADS works	Correlating citizen complaints with algorithmically flagged high-risk works

11.1 Data Acquisition Approach

- Phase 1 (Hackathon): use the publicly reachable Empowered Indian API plus the provided allocation dataset; where fields are missing (sanction dates, contractor names, geo-coordinates), generate clearly labelled synthetic augmentation data with documented, realistic fraud-injection rates (see Section 13) so that detector logic can be fully demonstrated without misrepresenting simulated data as real.
- Phase 2 (Pilot): formally request eSAKSHI bulk-export or read-access API credentials from MoSPI/NIC for one willing pilot district, to validate detectors against genuine Tier-1 fields.

- Phase 3 (Scale): integrate PFMS payment data, Bhuvan geospatial layers, and Census demographic data through official NIC data-sharing agreements, enabling the full Module A–E capability set nationally.

12. AI/ML Methodology — Detailed Detector Design

12.1 NLP Prohibited-Category Detector

- Approach: bilingual (English/Hindi) keyword and phrase dictionary (temple, mandir, masjid, church, gurdwara, mandap, satsang, ashram, math, shrine, and their Hindi/regional equivalents) as a fast first-pass rule engine, upgradeable to a fine-tuned multilingual text classifier (e.g., IndicBERT) trained on CAG-documented prohibited-work examples for higher recall on paraphrased descriptions.
- Validation: precision/recall measured against a manually labelled sample of confirmed prohibited vs. permitted works, drawn from CAG audit paragraphs.
- Output: binary flag + matched keyword span + confidence score, feeding into the composite risk score.

12.2 Geographic Mismatch Detector

- Approach: for each work, compare the recorded location/district field against the MP's official constituency boundary (sourced from ECI shapefiles once available); classify as Confirmed Mismatch, Boundary-Ambiguous (requires verification), or Consistent.
- Important design safeguard: as demonstrated in Section 7.5, this detector must never auto-escalate a mismatch as fraud without first checking whether the constituency genuinely spans multiple administrative districts — the system defaults to a verification-required state, never an accusation state, until boundary data confirms the finding.

12.3 Cost-Cluster and Cost-Overrun Detector

- Approach (Phase 1, cost-cluster only): frequency analysis of (work_type, cost) tuples per MP/district; flag clusters where identical cost recurs above a configurable threshold (e.g., >3 occurrences) without site-specific variation.
- Approach (Phase 2, full cost-overrun, once estimated_cost + SOR data available): Isolation Forest anomaly detection on standardised cost-per-unit features (cost per square metre, per kilometre, per unit), benchmarked against district/state Schedule-of-Rates baselines; regression models to forecast expected cost ranges by work type and location.

12.4 Delay Prediction Model

- Approach: supervised classification/regression (gradient-boosted trees, e.g., XGBoost/LightGBM) trained on historical sanction-to-completion time deltas, using features such as district administrative capacity, work type, season, implementing-agency track record, and fund-release lag.
- Output: probability of exceeding the 45-day sanction window or the broader completion deadline, generated at the point of sanction — enabling proactive intervention rather than after-the-fact audit.

12.5 Contractor Network / Fraud-Ring Analysis

- Approach: construct a bipartite-to-multipartite graph of Implementing Agency → Contractor → Bank Account → Work nodes using NetworkX; compute centrality metrics to identify contractors with disproportionate work concentration, and identity-resolution heuristics (fuzzy name matching, shared PAN/bank-account clustering) to detect blacklisted contractors reappearing under altered names.

12.6 Ghost-Work / Photo-Forensic Verification

- Approach: perceptual image hashing and similarity search to detect reused/duplicate photographs across unrelated works; EXIF GPS metadata extraction and reverse-geocoding to confirm the photo's capture location matches the claimed work site within a tolerance radius; flag stock-photo or AI-generated image signatures where detectable.

12.7 Composite Risk Score Engine

- Approach: a weighted, documented aggregation of all individual detector outputs into a single 0–100 score, with SHAP-based per-feature attribution so any stakeholder can see exactly why a work scored as it did.
- Governance: weights are configurable and versioned, not hard-coded black-box constants — every score is reproducible against the exact model version and input snapshot that produced it, supporting formal audit and appeal.
- Escalation policy: scores above a configurable threshold (e.g., 80/100) are routed to the District Authority / Ministry review queue; scores are never used to auto-block payments or auto-punish officials without human review.

13. Implementation Roadmap

13.1 Phase 1 — Hackathon MVP (36 Hours)

Since bulk, unrestricted real-time access to eSAKSHI is not publicly available, the hackathon build uses a hybrid data strategy: real data from the Empowered Indian API (as demonstrated throughout this document) combined with a clearly labelled synthetic augmentation layer for fields that are genuinely absent (sanction dates, contractor identity, geo-coordinates), so the full detector suite can be demonstrated end-to-end without misrepresenting simulated data as authentic.

1. Ingest real completed-works data via the Empowered Indian API with full pagination handling (demonstrated: 148 works across 8 pages for Kandhamal alone).
2. Generate a labelled synthetic augmentation set (~5,000 records: 542 MPs × ~10 works) with injected, documented fraud patterns — 5% duplicate works, 3% inflated cost estimates, 2% ghost-work markers, 10% delayed beyond 45 days — explicitly disclosed as synthetic in the demo.
3. Build and validate the two highest-value, data-available detectors first: the NLP prohibited-category detector and the cost-cluster detector (both already proven functional on real data in Section 7).
4. Build one predictive model (Delay Predictor) on the synthetic-augmented dataset to demonstrate the Module B forecasting capability.
5. Build three role-based dashboard views: MP View, District Authority View, Ministry View, each consuming the same underlying risk-scored API.
6. Prepare one live, narrated demo: run the anomaly engine against the real Kandhamal sample on stage, showing the exact 55% religious-work flag rate and 70% geo-mismatch flag rate documented in this report.

13.2 Phase 2 — Pilot (Months 1–3 Post-SIH)

- Approach a single, digitally mature District Authority (e.g., a district with strong eSAKSHI adoption) for a formal data-sharing MoU.
- Obtain read-only eSAKSHI API access or scheduled CSV exports covering Tier-1 fields (sanction dates, estimated costs, contractor identity).
- Deploy the anomaly-detection engine against genuine district data and validate algorithmic flags against physical inspection outcomes conducted by the District Authority.
- Iterate detector thresholds and weights based on real false-positive/false-negative rates observed during the pilot.

13.3 Phase 3 — Scale to National Deployment (Months 4–12)

- Present validated pilot results (with quantified inspection-efficiency gains) to MoSPI's MPLADS Division for formal adoption sign-off.
- Integrate directly with eSAKSHI and PFMS through NIC-brokered official data-sharing agreements, replacing the third-party API dependency entirely.
- Onboard Census, Bhuvan, and cross-scheme (MGNREGA/PMAY) data sources to unlock the full Tier-2/Tier-3 detector suite.
- Roll out district-by-district, prioritising districts with historically high CAG-flagged irregularity rates, scaling to all districts nationally.

13.4 Roadmap Timeline Summary

Phase	Duration	Primary Deliverable
Phase 1 — Hackathon MVP	36 hours	Working demo on real API data + synthetic augmentation; 3 role-based dashboards; live anomaly detection on Kandhamal sample
Phase 2 — Pilot	Months 1–3	Validated detection accuracy against physical inspection outcomes in one district
Phase 3 — National Scale	Months 4–12	Full eSAKSHI/PFMS/Census/Bhuvan integration; phased national rollout

14. Expected Impact and Outcome Metrics

The following targets are proposed as directional goals for a mature deployment, benchmarked against the current, documented baseline. They are framed as targets to validate through the phased pilot in Section 13, not guaranteed outcomes — consistent with this document's research-integrity standard.

Metric	Current Baseline	MPLADS-SATHI Target
Average fund utilisation	~33.9% (national)	>70–75% via predictive nudges and proactive fund-lapse warnings

Metric	Current Baseline	MPLADS-SATHI Target
Works with sanction delay beyond 45 days	~57% of sampled works	<10–15% via automated 30-day/45-day escalation alerts
Cost-overrun detection timing	Reactive — detected years later via CAG audit	Proactive — flagged at sanction/estimate stage, before disbursal
Ghost/duplicate work detection	Manual, ad hoc (e.g., the ₹161 crore undocumented-expenditure finding)	AI-flagged within 24–48 hours of data upload, pending photo/geo integration
Physical inspection targeting	Random 10% sample	Risk-prioritised 10% sample — same inspection capacity, redirected to highest-risk works
Compliance reporting cadence	Manual, quarterly	Real-time, continuously updated dashboards
Citizen grievance handling	Unstructured, largely untracked	Geo-tagged, AI-triaged, and cross-referenced against algorithmic risk flags

14.1 Demonstrated Proof Points (Not Projected — Already Observed)

Unlike the forward-looking targets above, the following are outputs already produced by applying this document's detection logic to the real 20-work Kandhamal sample, and are cited here as evidence the approach works on genuine data, not merely in theory:

- 11 of 20 works (55%) flagged for prohibited religious-category keywords.
- 14 of 20 works (70%) flagged for location/district field inconsistency, correctly labelled as verification-required rather than confirmed fraud.
- 4 solar-light installations flagged for identical, non-varying cost.
- 20 of 20 works (100%) flagged for zero beneficiary reporting — a systemic data-quality gap worth ministry attention independent of any fraud question.
- Extrapolating this sample's flag rate across an estimated ~54,000 works nationwide (542 MPs × ~100 works each) would direct India's fixed 10% manual-inspection capacity (~5,400 works) toward the highest-risk subset rather than a random sample — the single largest efficiency gain this solution offers even before any new data source is integrated.

15. Honest Capability and Limitation Assessment

A defensible SIH submission must be explicit about what the proposed system can and cannot do at each stage of data availability. This section is deliberately blunt.

15.1 What Can Be Built Today (Tier-1-Independent)

- NLP keyword flagging for prohibited/religious works — proven functional on real data.
- Geographic mismatch detection (location text vs. constituency district) — proven functional, correctly caveated pending ECI boundary confirmation.
- Identical-cost cluster analysis — proven functional on real data.
- Category–description consistency checking via NLP — proven functional on real data.
- Zero-beneficiary / basic data-quality flagging — proven functional on real data.
- Sector/portfolio concentration and bias analysis against state/national benchmarks.
- Allocation-fairness analytics across the full 542-MP dataset (state, category, deviation).

15.2 What Cannot Be Built Without Additional Data

- Delay prediction — requires sanction_date and work_order_date, currently absent from the observed API.
- True cost-overrun detection — requires estimated_cost alongside actual cost, currently absent.
- Contractor fraud-network analysis — requires contractor/implementing-agency identity fields, currently absent.
- Ghost-work photographic verification — requires geo-tagged completion photographs, currently absent from the sampled endpoint.
- Payment-velocity anomaly detection — requires PFMS payment-history integration.
- Automated SC/ST earmarking verification — requires village-level Census geo-join, not yet integrated.
- Cross-scheme duplicate-funding detection — requires MGNREGA/PMAY/Smart Cities location data, not yet integrated.

15.3 Risk Factors and Mitigations

Risk	Impact	Mitigation
False positives damaging an MP's or agency's reputation	High — public trust and political sensitivity	Confidence-graded flags (Confirmed / Verification-Required / Data-Quality), mandatory human review before any public escalation, full SHAP explainability
Data access delays from government stakeholders	Medium — could stall Phase 2/3 timeline	Design system to remain useful on Tier-1-independent detectors alone; pursue pilot with a single willing district rather than waiting for national access
Third-party API (Empowered Indian) rate limits, structure changes, or discontinuation	Medium — affects hackathon/demo continuity	Cache retrieved data locally; design ingestion layer with an adapter pattern so switching to official eSAKSHI API requires no architectural change

Risk	Impact	Mitigation
Model bias from unrepresentative training data	Medium — could systematically over-flag certain states/parties	Regular fairness audits across state, party, and reservation-category dimensions; transparent, published weighting methodology
Over-reliance on automated scores by under-resourced District Authorities	Medium — risk of rubber-stamping AI output without independent judgement	Explicit "AI-assisted, not AI-decided" governance policy; mandatory documented human rationale for any action taken on a flagged work

16. Consolidated Comparison with Existing Solutions

This section consolidates the system-level comparison from Section 5.2 with the concrete evidence from Section 7 to make explicit why MPLADS-SATHI represents a genuine capability addition rather than a duplicate of existing tools.

Existing System	What MPLADS-SATHI Adds Beyond It
eSAKSHI	Adds a full analytical intelligence layer (NLP, geospatial, statistical, predictive, graph, computer-vision detectors) on top of eSAKSHI's workflow records, plus automated compliance auditing against the 2023 Guidelines that eSAKSHI does not itself enforce.
Empowered Indian Dashboard	Converts a read-only public visualisation into an actionable, role-based decision-support system with risk scoring, alerting, and inspection-priority queues — while still using Empowered Indian's API as a legitimate, demonstrated real-data source.
PFMS	Adds fraud-relevant interpretation (contractor networks, payment-velocity anomalies) on top of PFMS's transaction-recording function, which PFMS itself does not attempt.
CAG post-facto audit process	Shifts detection from reactive, multi-year-lagged audit sampling to continuous, near-real-time, risk-prioritised monitoring — without replacing CAG's constitutional audit role, which MPLADS-SATHI's outputs can directly feed into.

17. Alignment with SIH 2026 Evaluation Criteria

17.1 Solves Real, Documented Pain Points

Every module in this proposal traces directly to a specific CAG finding, Parliament Q&A record, or a concrete anomaly demonstrated on real data in Section 7 — not a generic "AI for governance" framing.

17.2 Builds on Existing Government Infrastructure

The Government has already invested in eSAKSHI and PFMS; this solution's overlay architecture is designed to be immediately administratively and politically feasible to pilot, since it requires no migration off existing systems.

17.3 Technically Grounded, Not Merely Aspirational

Unlike a purely conceptual pitch, this proposal is anchored in a genuine, reproducible forensic analysis of real government-linked data (Sections 6–7), with an explicit, honest gap analysis (Sections 8, 15) — a level of rigour that differentiates a credible engineering submission from a slideware pitch.

17.4 Multi-Stakeholder, Nationally Scalable Impact

- MPs: better constituency management and transparent, defensible performance visibility.
- District Authorities: prioritised inspection queues and reduced manual tracking burden.
- State Nodal Authorities: automated reporting for state-level monitoring committees.

- Ministry: Parliament- and CAG-ready analytics, generated continuously rather than reconstructed retrospectively.
- Citizens: transparency, participatory oversight, and increased public trust in fund utilisation.

18. Team Composition and Required Skills

Delivering the full MPLADS-SATHI scope — even at hackathon MVP level — requires a genuinely cross-functional team. The following role breakdown reflects the skill sets actually exercised in building the modules described in Sections 9–12.

Role	Core Responsibility	Key Skills
Data Engineer	API ingestion, pagination handling, ETL pipeline, PostgreSQL/PostGIS schema design	Python, SQL, Airflow, REST API integration, geospatial data handling
ML/NLP Engineer	Anomaly detectors (NLP, Isolation Forest, clustering), model evaluation	scikit-learn, spaCy/Transformers, statistics, feature engineering, SHAP
Backend Developer	FastAPI services, risk-score aggregation engine, auth/RBAC	Python, FastAPI, Pydantic, API design, security fundamentals
Frontend Developer	Role-based React/Next.js dashboards, data visualisation	React, TypeScript, Recharts/D3.js, Leaflet.js, Tailwind CSS
Domain/Policy Researcher	MPLADS Guidelines interpretation, CAG audit synthesis, compliance-rule translation into detector logic	Public-policy research, government-document analysis, careful sourcing and fact-checking
Presenter / Product Lead	Pitch narrative, demo flow, jury Q&A defence, honest capability framing	Communication, storytelling grounded in evidence, technical fluency across all modules

19. Conclusion

MPLADS involves an annual public-fund corpus of roughly ₹8,318 crore across 542 Members of Parliament, executed through thousands of works nationwide. The Government has already digitised the scheme's workflow through eSAKSHI and its payment rails through PFMS — but neither system, nor the third-party Empowered Indian dashboard, currently applies any analytical intelligence over the data it holds. CAG audits have repeatedly and consistently documented the resulting cost: under-utilised funds, systemic sanction delays, inflated cost estimates, undocumented expenditure, and inadmissible works.

This document has shown, using genuine data rather than a hypothetical scenario, that even a first-pass rule-based detection engine — applied to just 20 real completed-works records from one constituency — surfaces concrete, actionable findings: a 55% prohibited-category flag rate, a 70% geographic-consistency flag rate requiring formal verification, template cost patterns, and a complete absence of beneficiary reporting. These are not manufactured examples; they are the direct output of the methodology described in Section 4, applied honestly and with appropriate confidence grading.

MPLADS-SATHI proposes to scale exactly this kind of evidence-grounded detection into a modular, explainable, multi-stakeholder platform — one that respects the existing government technology investment, is honest about what current data can and cannot support (Section 15), and is architected with a clear, phased path from a 36-hour hackathon demonstration to a nationally deployed, NIC-hosted production system. The problem is real, well-documented, and consequential for public trust in how India's elected representatives' development funds are spent. The data gap is real and precisely mapped in this document. The opportunity for a genuinely useful, technically sound AI solution is equally real — and this proposal is built to deliver it responsibly.

20. Sources and References

This document draws on the following categories of source material. Where a specific figure is cited in the body of this report, it originates from one of these categories as identified in context.

- Official problem statement text as issued for SIH 2026, MPLADS AI-monitoring theme (Government of India).
- MPLADS Guidelines 2023 — scheme rules on sanction timelines, earmarking, inspection quotas, and prohibited categories.
- eSAKSHI portal, Ministry of Statistics and Programme Implementation — mplads.mospi.gov.in/digigov/dashboard.html.
- Empowered Indian transparency platform and public developer API — empoweredindian.in/mplads and api.empoweredindian.in.
- eSAKSHI citizen-services description via Vikaspedia — en.vikaspedia.in (e-governance / online citizen services / MPLADS–eSAKSHI).
- Comptroller and Auditor General (CAG) of India — MPLADS performance audit findings on cost overruns, undocumented expenditure, and inadmissible works.
- Lok Sabha Starred Question records (2026) on MPLADS delays and accountability mechanisms.
- Primary dataset: "Allocated Limit for Hon'ble MPs" register (542 MPs, provided source data).
- Primary API sample: live query response from api.empoweredindian.in for Kandhamal constituency, Odisha (Lok Sabha, 18th term), retrieved and analysed for this document.
- Union Cabinet approval of MPLADS continuation through FY 2030-31, including third-party evaluation and Prabhari Officer monitoring provisions.

Appendix A — Sample Allocation Data Extract (MP Allocated-Limit Register)

The table below reproduces a representative extract of the cleaned 542-MP allocation dataset analysed in Section 6, illustrating the standard, above-standard, and below-standard allocation categories referenced throughout this document.

State	MP Name	Constituency	Allocated Amount
Telangana	Eatala Rajender	Malkajgiri	₹32.75 Cr
West Bengal	SK Nurul Islam	Basirhat	₹4.90 Cr
Assam	Pradyut Bordoloi	—	₹9.80 Cr
Kerala	Priyanka Gandhi Vadra	Wayanad	₹12.25 Cr
Bihar	(Standard allocation example)	—	₹14.70 Cr
Puducherry	(State average shown)	—	₹20.94 Cr avg

Appendix B — Sample Work-Level Data Extract (Kandhamal, Odisha)

The table below reproduces a representative extract of the 20-work Kandhamal sample analysed in Section 7, showing the exact fields returned by the live API query used for this research.

Work ID	Work Description	Cost (₹)	Category	Flag Raised
179092	Solar street light at boundary of Daberi Ashram School	5,00,000	Normal/Others	Identical-cost cluster
178816	Completion of incomplete Cultural Mandap at Adivasi Sahi, Nuagaon	1,50,000	Normal/Others	Religious keyword
178765	Boundary wall of Mukteswar Jew community centre, Daspalla NAC	1,00,000	Normal/Others	Religious keyword + geo mismatch
179089	Solar street light at boundary of Badabanga Ashram School	5,00,000	Normal/Others	Identical-cost cluster
178842	Boundary wall of Kaima Anganwadi House	1,00,000	Normal/Others	Geo mismatch (verification-required)

Appendix C — Glossary of Key Terms

Term	Definition
MPLADS	Members of Parliament Local Area Development Scheme — a Central Sector Scheme enabling MPs to recommend local development works.
eSAKSHI	The official MoSPI digital workflow portal for MPLADS recommendation, sanction, and execution tracking.
PFMS	Public Financial Management System — the Government's central payment and fund-tracking infrastructure.
CAG	Comptroller and Auditor General of India — the constitutional audit authority for public expenditure.
District Authority (DA)	The district-level administrative authority responsible for sanctioning and monitoring MPLADS works.
Implementing Agency	The government body or registered agency executing a sanctioned MPLADS work.
SHAP	SHapley Additive exPlanations — a technique for attributing a machine-learning model's output to individual input features.
Isolation Forest	An unsupervised machine-learning algorithm used to detect statistical outliers/anomalies in numeric data.
PostGIS	A spatial-database extension for PostgreSQL enabling geographic queries and analysis.
Risk Score	A composite, explainable 0–100 score assigned to a work, aggregating all applicable anomaly-detector outputs.